

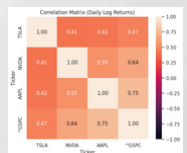


The Tech Domino Effect: Volatility Transmission from Mega-Caps to the S&P 500

Vo Thi Quynh Tram, Lai Chiu Lok, Huei-Wen Teng

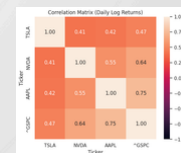
Motivation& Introduction

- Rising dominance of mega-cap tech firms in the market.
- Potential volatility spillovers affecting overall market movements.
- Need to assess systemic risk from concentrated market influence.



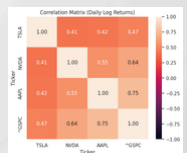
Outline

1. Motivation & Introduction ✓
2. Data & Benchmark method
3. Study plan & results



Data Preprocess_Data Overview

- ▣ Source : Yahoo Finance (Adjusted Close)
- ▣ Period : 2015–2024 (Daily)
- ▣ Target Assets : Apple (AAPL), Nvidia (NVDA), Tesla (TSLA),
S&P 500 (Benchmark)
- ▣ Transformation : Log returns → Rolling Volatility (annualized)



Data Preprocess_Variables

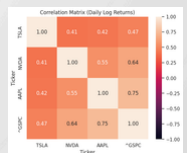
Variable	Definition	Role
RV_SPX	Rolling Volatility of S&P 500	Dependent
RET_AAPL, RET_NVDA, RET_TSLA	Log Returns of tech firms	Independent
RV_AAPL, RV_NVDA, RV_TSLA	Rolling Vol of tech firms	Spillover Drivers



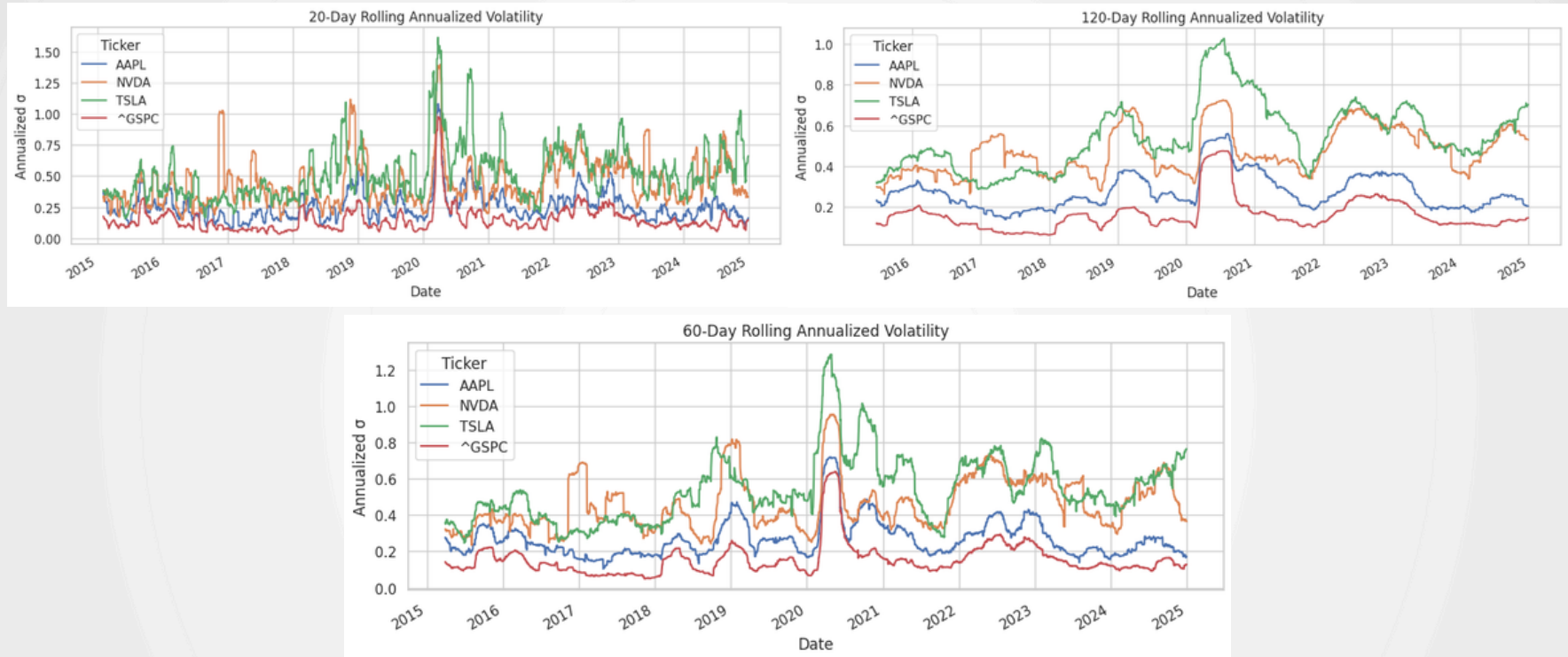
Data Preprocess_Descriptive Statistics

Ticker	count	mean	std	min	25%	50%	75%	max	Skew	Kurtosis	JB_pvalue
AAPL	2514	0.000930	0.017934	-0.137708	-0.007355	0.001000	0.010093	0.113158	-0.201228	5.457399	0
NVDA	2514	0.002248	0.030378	-0.207712	-0.012514	0.002648	0.017472	0.260876	0.208734	6.829886	0
TSLA	2514	0.001333	0.035911	-0.236518	-0.016292	0.001264	0.019073	0.198187	-0.066409	4.410539	0
SPX	2514	0.000419	0.011272	-0.127652	-0.003780	0.000639	0.005740	0.089683	-0.810830	15.750819	0

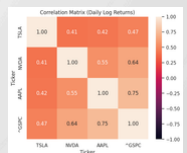
- ▣ All assets: high variance (Std highest for TSLA)
- ▣ Evidence of non-normality:
 - ▶ Positive/negative skew → asymmetric risk
 - ▶ High kurtosis → heavy tails
 - ▶ JB-test → rejects normality ($p = 0$)



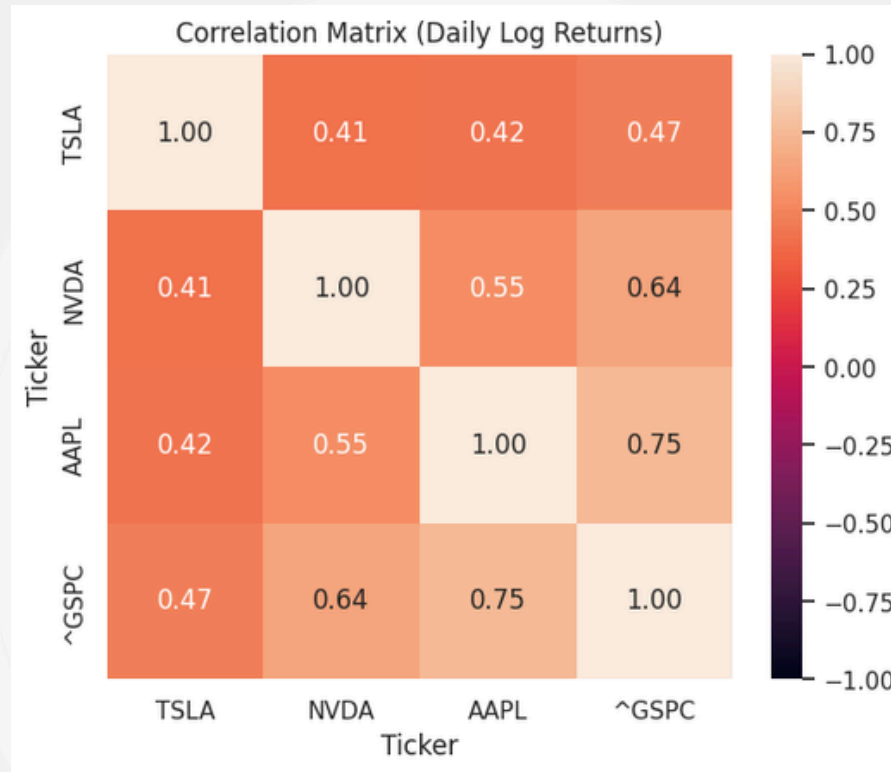
Data Preprocess_Rolling Volatility



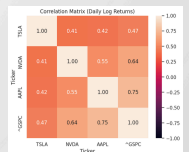
- TSLA & NVDA show sharp volatility spikes (2020–2021)
- SPX volatility rises after mega-cap spikes → Possible contagion
- AAPL = moderate, smoothed volatility (market stabilizer)



Data Preprocess_Correlation & Co-Movement

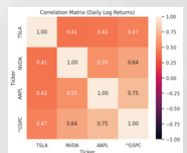


- AAPL highly correlated with SPX (~0.75)
- TSLA shows episodic decoupling (idiosyncratic risk)
- NVDA moderately correlated, rising in AI boom
- Conclusion: Strong integration but not perfect coupling → transmission, not redundancy.



Data Preprocess_Key Findings (Volatility Transmission)

- Evidence of fat tails, extreme risk, and high Beta in mega-caps
- S&P500 volatility spikes lag mega-cap volatility surges
- Tech sector acts as a volatility leader, not follower



Benchmark Model

Linear regression

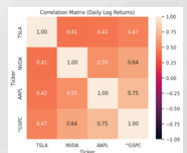
OLS linear regression using full 15 predictors

Benchmark equation:

$$SPX_RV60_t = \beta_0 + \sum_j (\beta_1 RV20_{j,t-1} + \beta_2 RV60_{j,t-1} + \beta_3 RV120_{j,t-1} + \beta_4 r_{j,t-1}) + \varepsilon_t$$

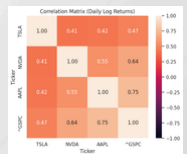
Captures:

- Endogenous SPX volatility persistence
- Exogenous spillovers from AAPL, NVDA, TSLA
- Provides baseline for evaluating **LASSO** and **nonlinear ML models**

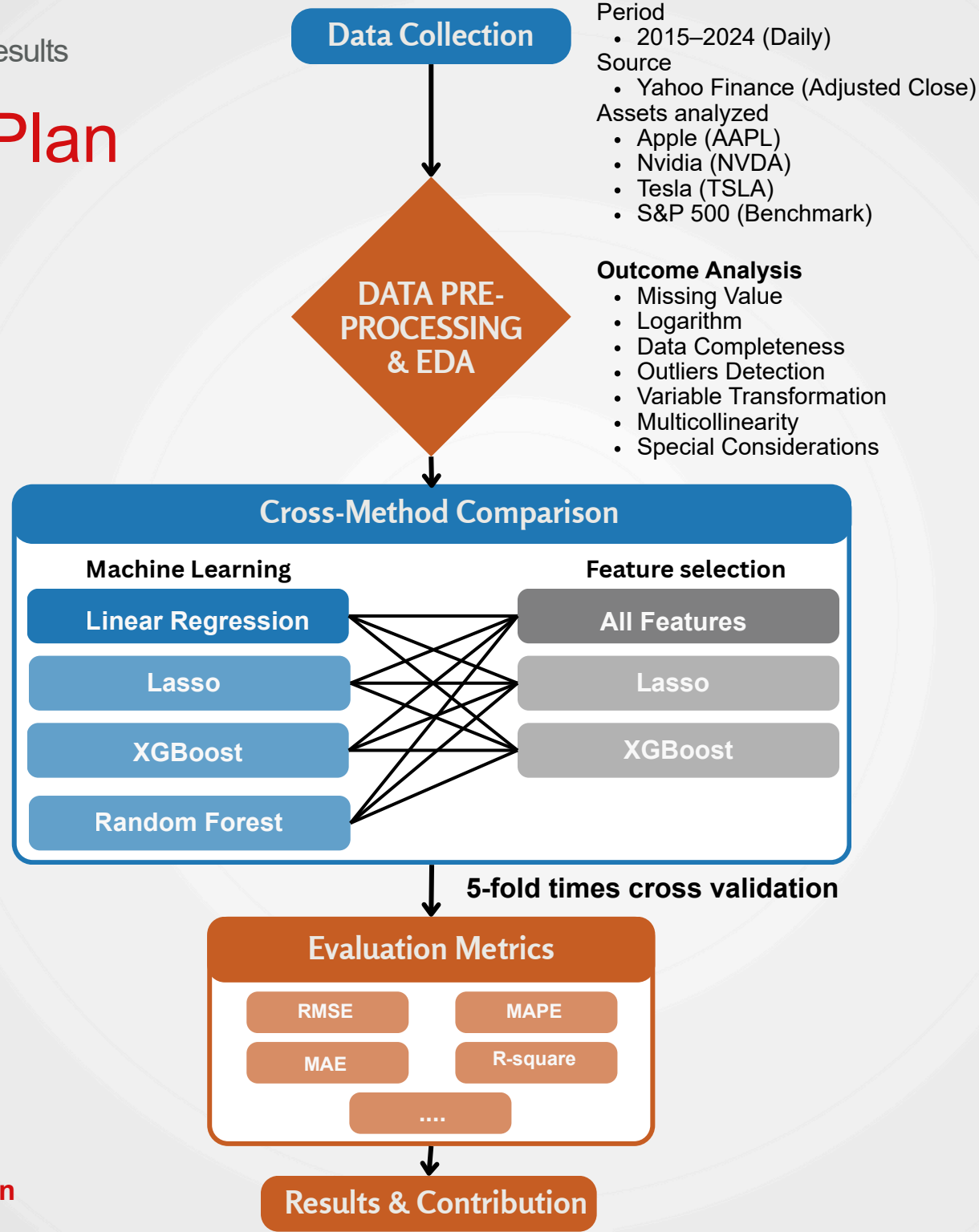


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Study Plan



Linear Regression



Methodology Overview

Objective:

- Apply a machine-learning-enhanced **HAR volatility framework** to study spillovers from **AAPL, NVDA, and TSLA to the S&P 500**
- Construct **daily log-returns** and **20-, 60-, 120-day realized volatilities** to capture short-, medium-, and long-term volatility persistence
- Use **SPX 60-day realized volatility (SPX_RV60)** as the dependent variable, a standard measure of medium-horizon market uncertainty.



$$\mathcal{X} = \{r_{j,t-1}, RV_{j,t-1}^{20}, RV_{j,t-1}^{60}, RV_{j,t-1}^{120}\}_{j \in \{SPX, AAPL, NVDA, TSLA\}}.$$

All predictors are one-day lagged

Final feature set includes 15 lagged return and volatility predictors



Methodology

$$\mathcal{X} = \{r_{j,t-1}, RV_{j,t-1}^{20}, RV_{j,t-1}^{60}, RV_{j,t-1}^{120}\}_{j \in \{SPX, AAPL, NVDA, TSLA\}}.$$

- **AllFS:** all 15 predictors are retained.
- **LassoFS:** predictors with non-zero LASSO coefficients selected via cross-validation over $\lambda \in \{10^{-6}, \dots, 10^{-1}\}$.
- **XGBFS:** top predictors identified using XGBoost feature importance rankings.



Five-Fold Rolling Time-Series CV

EVALUATION METRICS

- RMSE
- MAE
- MAPE
- Median Absolute Error
- R-square
- Variance
- Correlation
- QLIKE



Machine-learning Model Performance

FS	Model	Features	RMSE	MAE	MAPE	MedAE	R^2	Corr	QLIKE
AllFS	LASSO	15	0.0263	0.0207	0.1598	0.0174	-0.1264	0.8454	0.0401
AllFS	LinearReg	15	0.0267	0.0212	0.1655	0.0177	-0.2553	0.8410	0.0423
AllFS	XGBoost	15	0.0379	0.0246	0.1698	0.0172	0.6525	0.8565	0.0261
AllFS	RandomForest	15	0.0430	0.0289	0.2034	0.0215	0.4941	0.7677	0.0281
LassoFS	LASSO	14	0.0263	0.0207	0.1598	0.0174	-0.1264	0.8454	0.0401
LassoFS	LinearReg	14	0.0267	0.0212	0.1654	0.0177	-0.2553	0.8410	0.0423
LassoFS	XGBoost	14	0.0373	0.0240	0.1675	0.0177	0.6483	0.8506	0.0281
LassoFS	RandomForest	14	0.0428	0.0289	0.2035	0.0215	0.4945	0.7670	0.0281
XGBFS	LASSO	10	0.0270	0.0220	0.1655	0.0181	-0.0220	0.8408	0.0464
XGBFS	LinearReg	10	0.0268	0.0221	0.1729	0.0181	-0.3763	0.8431	0.0518
XGBFS	XGBoost	10	0.0374	0.0249	0.1712	0.0173	0.6622	0.8484	0.0207
XGBFS	RandomForest	10	0.0424	0.0284	0.2039	0.0212	0.5193	0.7642	0.0271

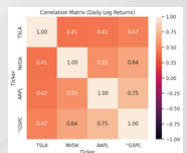
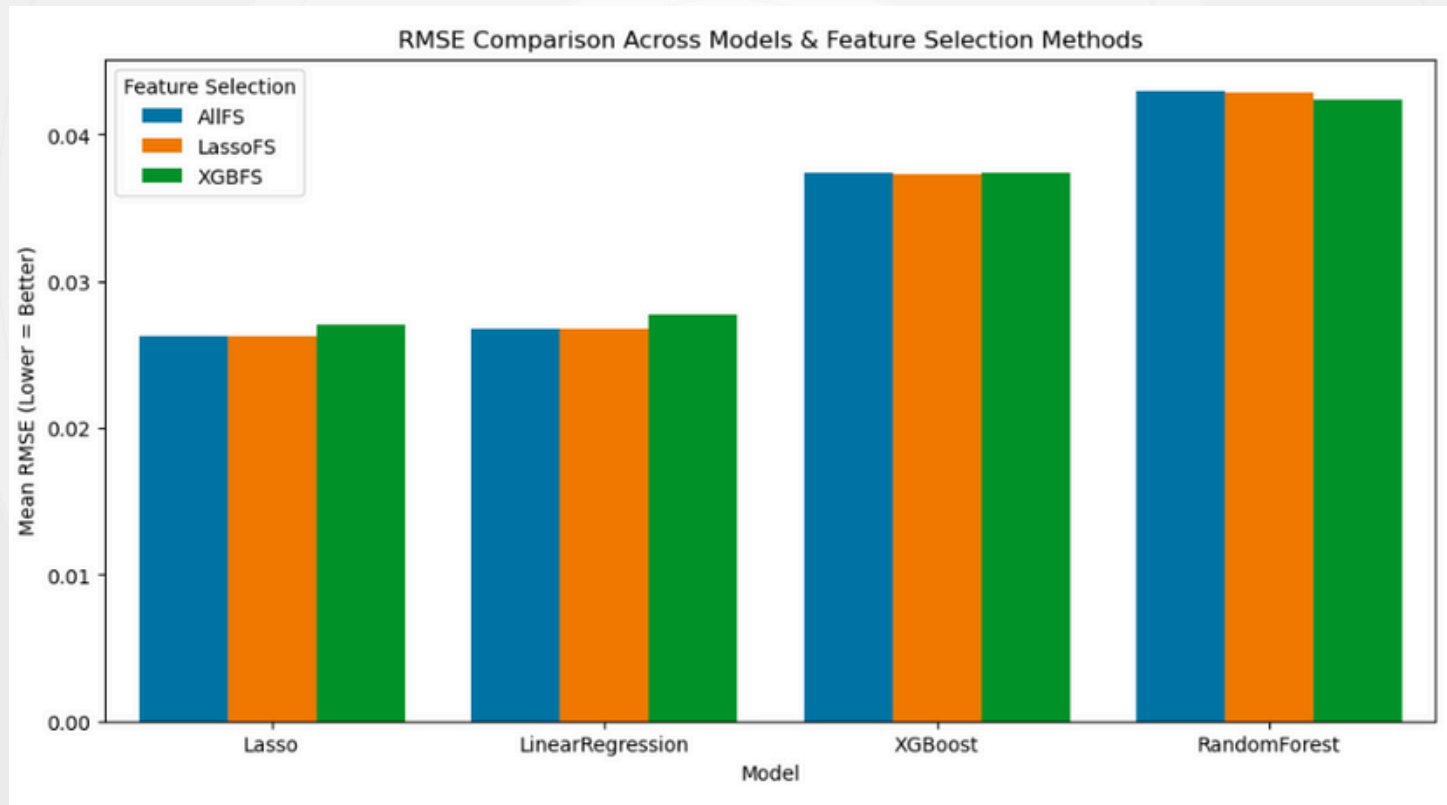
General Observations

- **LASSO** → Best RMSE & MAE (most accurate forecasts)
- **XGBoost** → Best R^2 , highest correlation, lowest QLIKE
- **Linear Regression** → Low error but negative R^2 → fails to explain variance
- **Random Forest** → Moderate performance; unstable in volatile regimes



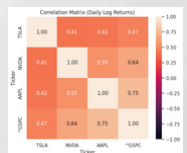
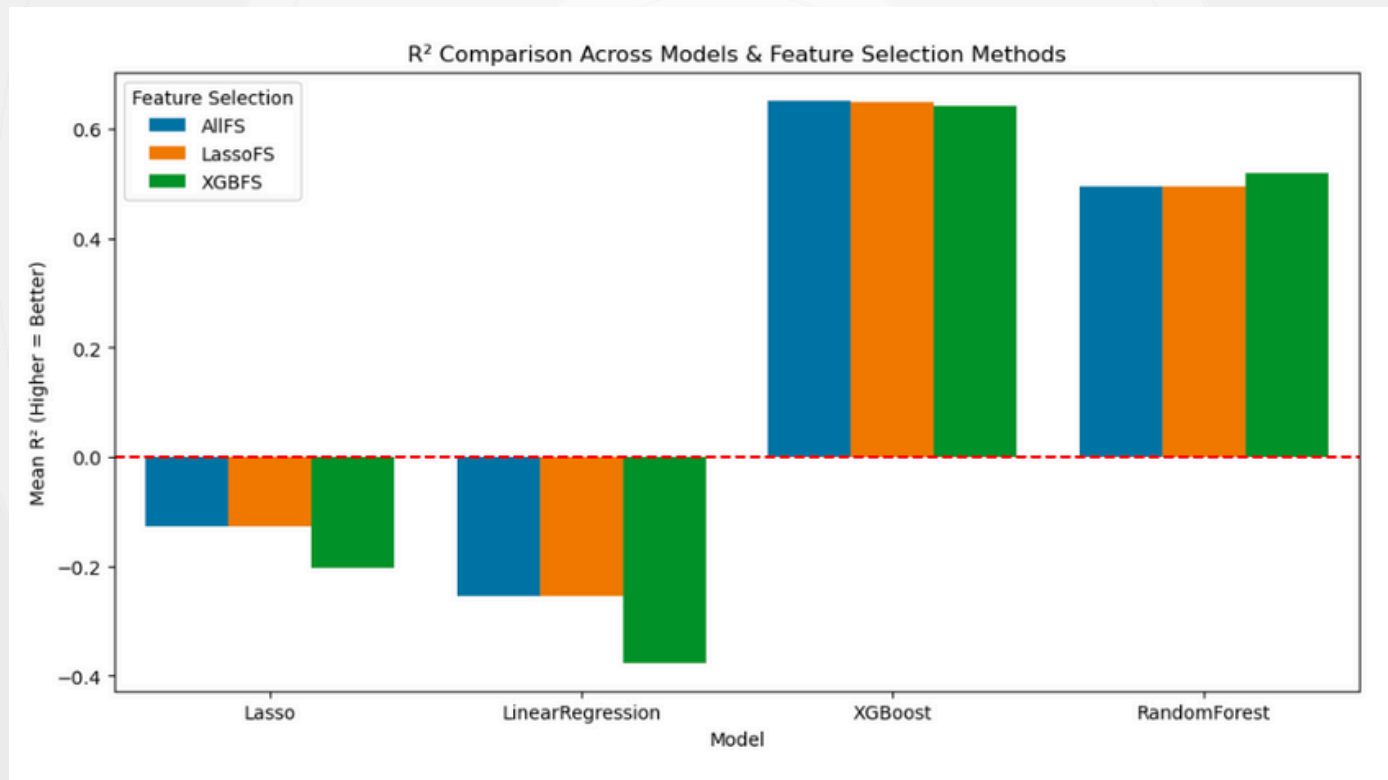
Machine-learning Model Performance

- **LASSO and Linear Regression have the lowest RMSE values**
- Feature selection (AllFS vs LassoFS vs XGBFS) has minimal impact on RMSE
- Random Forest performs worst across all feature-selection methods



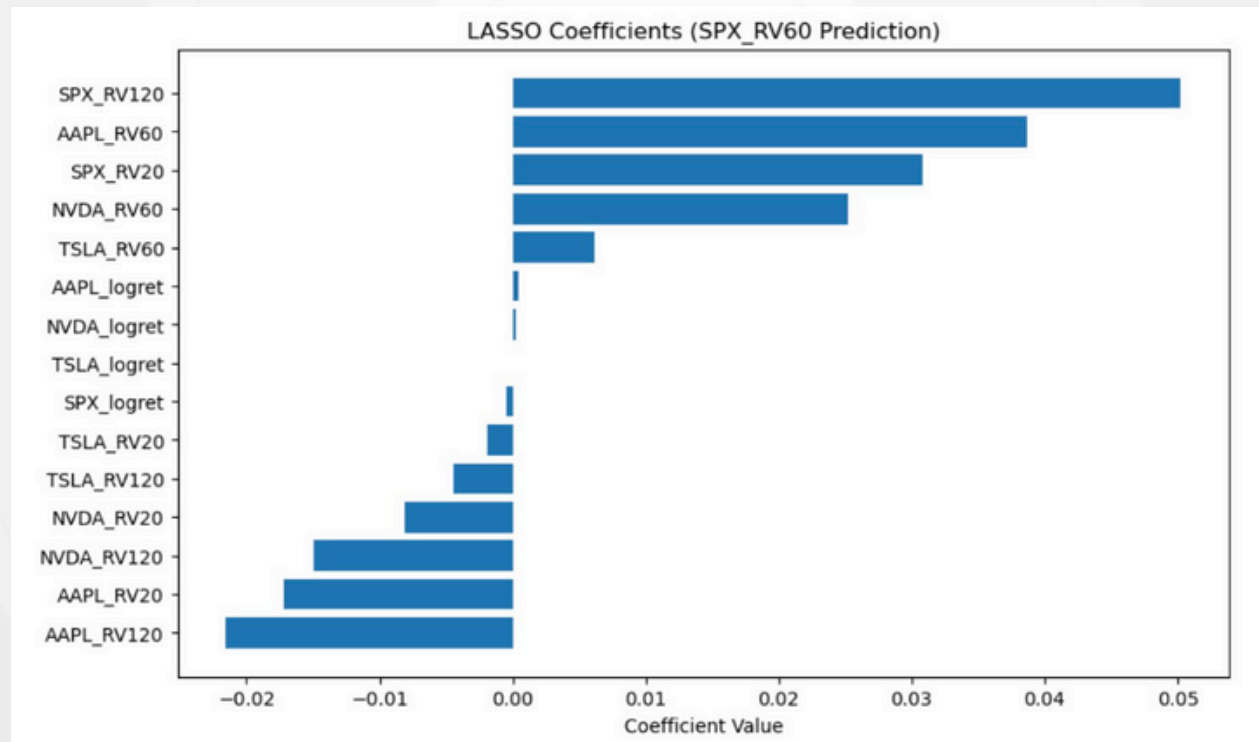
Machine-learning Model Performance

- **XGBoost has moderate RMSE—better than Random Forest, worse than linear models**
- **LASSO slightly outperforms Linear Regression**



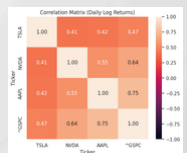
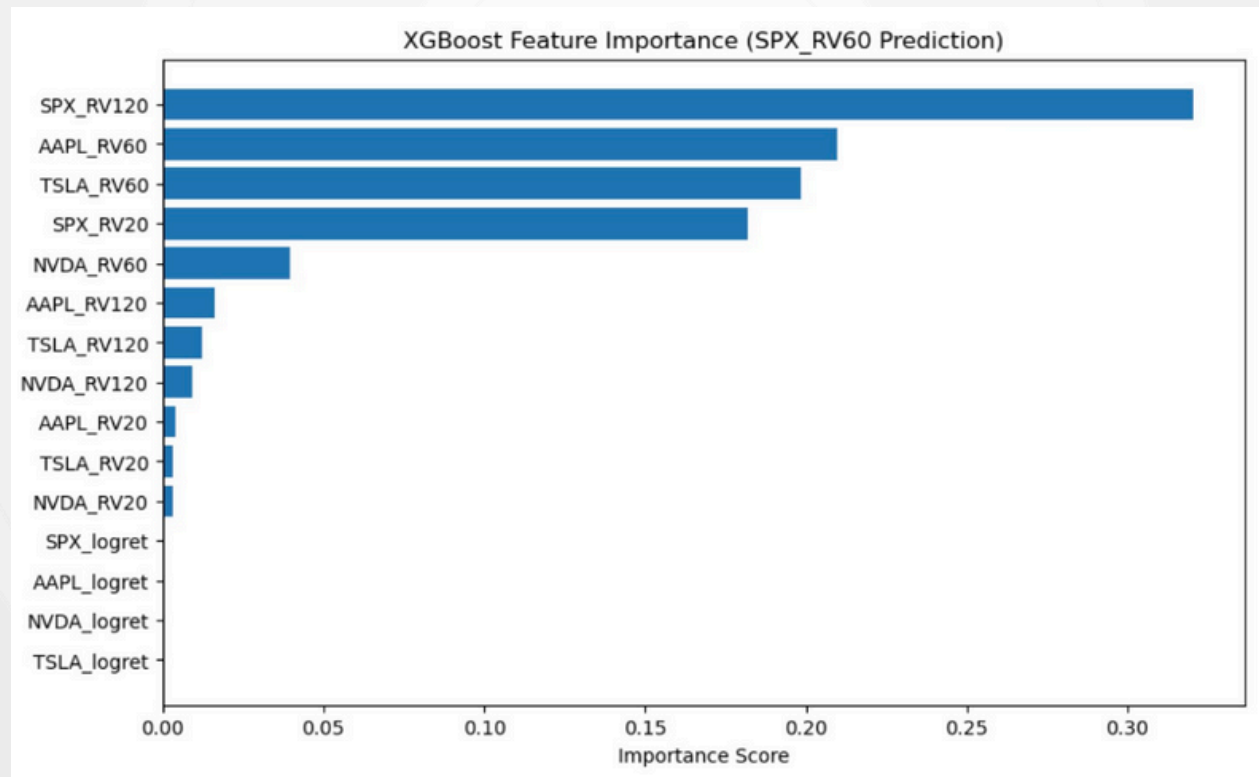
Results

- **Nvidia and Tesla show the strongest spillover effects** on S&P 500 volatility across all models and feature-selection methods.
- Their **medium- and long-horizon volatilities (RV60, RV120)** consistently rank as the most influential predictors in both LASSO coefficients and XGBoost feature importance.
- **Apple's volatility plays a weaker and less consistent role**



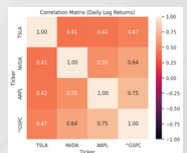
Results

- **Nonlinear dynamics dominate:**
 - OLS and LASSO achieve low RMSE but produce **negative R^2** , failing to explain actual volatility movements.
 - **XGBoost achieves the strongest explanatory power** (highest R^2 , correlation, lowest QLIKE), capturing nonlinear and asymmetric transmission patterns.
- Random Forest performs moderately but shows higher instability and weaker prediction accuracy.



Key insight & Contribution

- ✓ A small set of mega-cap tech firms exerts disproportionate influence on S&P 500 volatility.
- ✓ Spillovers intensify during crises, showing strong state-dependent effects.
- ✓ The HAR–ML framework captures both linear persistence and nonlinear amplification.
- ✓ ML models uncover nonlinear spillovers that linear models miss.
- ✓ Monitoring NVDA and TSLA volatility is critical for market risk forecasting.



Thank you

